SONIC (SOUNDSENSE)

Project Review 1

An on-device sound awareness system for Deaf and hard-of-hearing users

Department of Computer Science and Engineering

Dayananda Sagar University | 7th Semester

Review scope: finalized problem statement, objectives, literature review, research gap, proposed system design, methodology, tools and datasets, work plan, and initial working-module demonstration plan.

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1. Review Scope and Project Position

Sonic is proposed as an accessible, privacy-oriented sound awareness application that converts important ambient sounds into tactile, visual, and local notification cues. The project is intended primarily for Deaf and hard-of-hearing users, for whom time-sensitive environmental sounds may otherwise be missed.

This Review 1 document separates the present software baseline from the remaining project work. The submitted project material describes a Flutter application architecture, a DSP and inference pipeline, persistence, alerting, BLE integration points, and tests. These constitute the current implementation baseline. Training and validation of the production model, evaluation with representative recordings and users, and real wearable deployment remain planned activities.

2. Finalized Problem Statement

Deaf and hard-of-hearing individuals can miss safety- and context-relevant environmental sounds such as smoke alarms, sirens, doorbells, glass breaking, phone ringing, appliance beeps, and a specific household sound. Existing solutions may focus on a single personalized cue, depend on cloud connectivity, or provide insufficient personalization for acoustically diverse homes. The problem is to design an on-device system that detects a defined set of environmental sounds and user-taught sound instances, then delivers timely, understandable, and non-auditory alerts while retaining user control over sensitivity and feedback.

3. Objectives

No.	Objective
O1	Develop an on-device ambient audio pipeline for 16 kHz mono PCM capture, real-time DSP, and sound-event inference.
O2	Provide a 20-class general environmental-sound layer organized by urgency.
O3	Enable a few-shot teaching workflow that forms a user-specific prototype from 2 to 5 recordings.
O4	Deliver accessible multimodal alerts through vibration patterns, local notifications, visual strobe/HUD, and a BLE haptic interface.
O5	Store taught sounds, alert history, sensitivity settings, and feedback locally in SQLite.
O6	Reduce repeat alerts with per-sound debounce and adapt prototype sensitivity after repeated negative feedback.
O7	Evaluate technical correctness, detection behavior, and usability in subsequent project phases.

4. Current Status at Review 1

Area	Current evidence in supplied material	Status for review
Application foundation	Flutter/Riverpod feature structure; UI screens; local persistence; alert, audio, ML and BLE service layers are described.	Current implementation baseline
DSP and matching	Ring buffer, 512-point FFT, 64-band log-mel processing, 128-D vectors, cosine similarity, and prototype averaging are described.	Current implementation baseline
General classifier	Native TFLite interface and an acoustic fallback are described; a production trained model is not evidenced.	Planned model integration/validation
Wearable operation	BLE scanning/pairing and GATT alert interface, including mock devices, are described.	Interface baseline; physical-device validation planned
Evidence	Supplied material reports 0 analyzer issues/warnings and named unit/E2E tests. No measured accuracy or user-study outcomes are supplied.	Verification baseline only

5. Literature Review and Base Paper Comparison

The project is informed by work on personalized sound recognition, edge audio classification, environmental sound datasets, and assistive wearable alerting. Lumename is the base paper and establishes the relevance of personalized on-device auditory detection with haptic-visual alerts [1]. ProtoSound further motivates personalized and scalable recognition for Deaf and hard-of-hearing users [2]. AHEAD-DS and YAMNet+ indicate the continued development of environmental sound recognition for hearing-device contexts [3]. ACDNet provides evidence that efficient deep-learning approaches are important on edge devices [4]. The wearable system reported by Nguyen et al. establishes a related real-time sound-detection direction for hearing-impaired users [5]. ESC-50, UrbanSound8K, and AudioSet are relevant dataset resources for model development and evaluation [6]-[8].

Dimension	Lumename (base paper) [1]	Sonic proposed extension
Primary task	Personalized detection for a single keyword/name	General environmental awareness plus personal household-sound matching
Recognition approach	MFCC + custom CNN from scratch	YAMNet-based transfer-learning direction for general classes; 128-D embedding/prototype matching for taught sounds
Personalization	Personalized keyword detection	Two-tier design: universal categories and few-shot personal fingerprints
Outputs	Haptic and visual alerts	Haptic patterns, local notifications, visual flash/HUD, and BLE haptic transmission
Scope	Name-calling awareness	Safety, urgency, appliance and household awareness

6. Research Gap and Project Contribution

The supplied literature and project rationale indicate a gap between single-cue personalization and a practical, broader environmental-awareness workflow. Sonic addresses this gap by combining a fixed set of general sound categories with local, few-shot prototypes for personally meaningful instances. The intended contribution is architectural: a privacy-preserving on-device pipeline that joins DSP, classifier/embedding inference, adaptive alert control, accessible output modes, and local audit/feedback data. This is a proposed project contribution; comparative performance, generalization, and user benefit must be established in future evaluation rather than assumed.

7. Proposed Architecture and System Design

Audio is captured as 16 kHz mono PCM and accumulated in a 1-second rolling buffer. The DSP layer applies a Hann-windowed 512-point radix-2 FFT and produces 64 log-mel bands from 50 Hz to 8 kHz with hop size 256. The inference layer is designed to yield a general class decision and a 128-dimensional embedding. Live embeddings are L2-normalized and compared with stored few-shot prototypes using cosine similarity. Confident events pass to the coordinator and alert service; alerts are debounced per sound for 10 seconds, logged locally, and delivered as tactile, visual, notification, and BLE outputs.

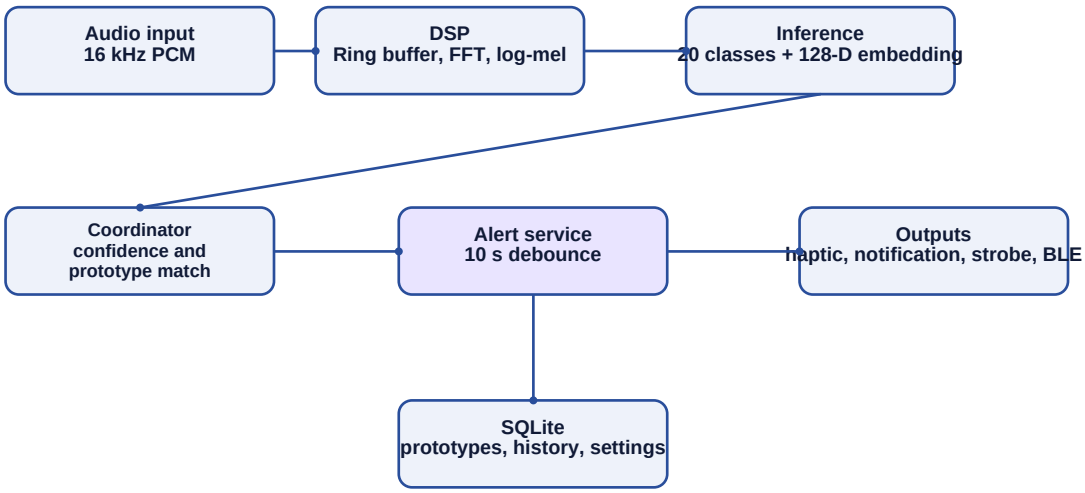

Figure 1. Proposed Sonic data and control flow.

8. Methodology

Phase	Method	Output / decision criterion
1. Requirements and sound taxonomy	Define 20 general classes across Critical, Urgent, Warning and Info tiers; identify personal-use cases.	Documented classes, urgency semantics, and alert mapping.
2. Dataset preparation	Use ESC-50, UrbanSound8K, and/or AudioSet where licensing and class alignment permit; create a held-out evaluation split.	Traceable dataset selection and label mapping.
3. General model development	Apply YAMNet-based transfer learning direction for the universal layer; export/validate a mobile-compatible inference artifact.	Model candidate with measured validation results.
4. Few-shot p ersonalization	Record 2-5 samples per taught sound; extract 128-D vectors; L2-normalize and average prototypes.	Stored prototype with a configurable similarity threshold.
5. Runtime integration	Connect DSP, classifier/embedding engine, coordinator, alert service, SQLite, and interface flows.	End-to-end on-device detection workflow.
6. Adaptive control	Apply 10-second per-sound debounce; record user feedback; raise threshold by 0.04 after 3 consecutive negative ratings.	Controlled reduction of repeat/false-positive alerts.
7. Evaluation	Measure detection behavior, latency, false alerts, persistence behavior, and accessibility/usability feedback.	Evidence-led final assessment.

9. Technical Design Details

Component	Design detail from supplied material
Audio and DSP	16,000-sample / 1-second window; signed PCM16 normalization; RMS/dB metrics; 512-point FFT; Hann window; 64 triangular mel filterbanks; log compression.
ML and matching	20-category classifier interface; 128-D L2-normalized embeddings; cosine similarity; average of 2-5 sample vectors; default matching threshold 0.85 (range 0.50-0.98).
Alert control	10-second debounce per sound key; category-specific vibration patterns; high-priority local notification; accessible visual flash overlay.
Adaptive feedback	History ratings track feedback; 3 consecutive thumbs-down events trigger +0.04 threshold adjustment.
Persistence	Drift/SQLite tables for taught sound metadata/embeddings, alert history, category settings, thresholds and feedback counters.
BLE and UI	BLE scanning, pairing and GATT haptic-send interface; onboarding, permission, live monitor, teach sound, history, settings, diagnostics and pairing screens.

10. Tools, Platforms and Datasets

Category	Tool / resource	Intended use
Application	Flutter, Dart, Riverpod	Cross-platform reactive mobile application structure.
Audio	PCM16 microphone stream; custom ring buffer, FFT and mel extractor	On-device audio capture and feature extraction.
ML runtime	TensorFlow Lite interface; YAMNet-based transfer-learning direction	Mobile inference integration; model training/export is planned.
Persistence	SQLite with Drift abstractions	Local prototypes, history, settings, and feedback.
Connectivity	flutter_blue_plus / BLE GATT interface	Wearable discovery/pairing and haptic alert transmission.
Notifications	flutter_local_notifications and vibration service	Local accessible alerts.
Datasets	ESC-50 [6], UrbanSound8K [7], AudioSet [8]	Candidate data sources for training/evaluation after class/license mapping.

11. Initial Working Module Demonstration Plan for the PPT

The review presentation should demonstrate only capabilities described in the supplied project material. It should clearly label simulated or fallback paths and should not portray a trained production model or physical wearable result unless the presenter has verified it live.

Demo step	What to show	Evidence / narration
1. Launch and consent	Onboarding and permission screens; on-device privacy explanation.	Establishes accessibility and local-processing intent.
2. Live monitor	Listening toggle, waveform, dB meter, status display, and alert overlay.	Shows the operational user interface.
3. DSP diagnostic	Log-mel spectrogram screen and category probability display.	Explain 16 kHz input, FFT 512, 64 mel bands, and 1-second rolling window.
4. Teach a sound	Record 2-5 samples, choose a label/icon, save prototype.	Explain 128-D vectors, prototype averaging, L2 normalization, and cosine similarity.
5. Detection and alert	Use sound simulator or controlled audio; show haptic/notification/strobe response.	Label as simulation where applicable; demonstrate 10-second debounce.
6. Feedback adaptation	Show history rating; explain threshold increase after 3 consecutive negative ratings.	Describe rule; do not claim false-positive reduction without measured data.
7. BLE path	Show pairing/mock wearable flow and tactile test trigger.	State whether device is mock or physical during presentation.

12. Work Plan

Stage	Planned activity	Reviewable output
Review 1	Finalize problem, objectives, gap, architecture, baseline software modules and demo script.	This review document and initial working-module demonstration.
Post Review 1	Prepare dataset mapping and preprocessing; train/transfer-learn universal classifier; select thresholds.	Reproducible data/model experiment record and preliminary metrics.
Mid project	Integrate validated model; test few-shot matching across controlled household sounds; refine alerts.	Functional end-to-end prototype with measured behavior.
Late project	BLE hardware validation, robustness testing, accessibility/usability feedback, and error analysis.	Evaluation results, limitations, and improvement log.
Final review	Consolidate implementation, findings, demo and documentation.	Final report and evidence-supported presentation.

13. Verification Baseline and Limitations

The supplied project material reports that **dart analyze lib** completed with 0 issues / 0 warnings. It also identifies tests for alert debounce, audio DSP, embedding operations, database CRUD/feedback state, and an end-to-end journey covering onboarding, permissions, teaching, detection target under 2 seconds, debounce, history, and adaptive tuning. These are valuable verification artifacts for code behavior. They are not equivalent to evidence of recognition accuracy, real-world robustness, hardware reliability, or accessibility effectiveness. Those claims require controlled data and user-centered evaluation in the planned phases.

The architecture also includes a fallback acoustic spectral analyzer when a compiled TFLite model is not present. For review accuracy, demonstrations that use this path should identify it as a fallback/simulation-oriented baseline, not as validation of a trained YAMNet-based model.

14. Conclusion

Sonic proposes a coherent two-tier sound awareness system: universal environmental categories provide broad coverage while few-shot personal prototypes support household relevance. The current project baseline concentrates on the application, DSP, matching, alerting, persistence, BLE interfaces, and verification scaffolding. The next project milestone is to produce measured model and system-evaluation evidence so that the intended benefits can be assessed rigorously.

References

[1] A. Dao and A. Dao, "Lumename: Wearable Device for Hearing Impaired with Personalized ML-Based Auditory Detection and Haptic-Visual Alerts," arXiv:2508.01576, 2025. https://arxiv.org/abs/2508.01576

[2] D. Jain, J. N. Gupta, A. N. Gnanapragasam, S. N. Patel, J. O. Wobbrock, and J. E. Froehlich, "ProtoSound: Personalized and Scalable Sound Recognition for Deaf and Hard-of-Hearing Users," CHI Conference on Human Factors in Computing Systems, 2022. https://arxiv.org/abs/2202.11134

[3] H. Zhong et al., "AHEAD-DS and YAMNet+: Environmental Sound Recognition for Hearing Devices," arXiv, 2025. https://arxiv.org/abs/2508.10360

[4] N. Gourisaria, C. R. Mohanty, and P. K. Biswal, "ACDNet: An Efficient Deep Learning Approach for Audio Classification on Edge Devices," arXiv:2103.03483, 2021. https://arxiv.org/abs/2103.03483

[5] T. N. Nguyen, S. V. Chhetri, D. Alahakoon, and D. Chua, "Sensor-Based Wearable System for Real-Time Detection of Important Sounds for Hearing-Impaired Users," Sensors, vol. 18, no. 8, 2018. https://www.mdpi.com/1424-8220/18/8/2503

[6] K. J. Piczak, "ESC-50: Dataset for Environmental Sound Classification," 2015. https://github.com/karolpiczak/ESC-50

[7] J. Salamon, C. Jacoby, and J. P. Bello, "UrbanSound8K: A Dataset of Urban Sound Recordings," ACM Multimedia, 2014. https://urbansounddataset.weebly.com/urbansound8k.html

[8] J. F. Gemmeke et al., "AudioSet: An Ontology and Human-Labeled Dataset for Audio Events," ICASSP, 2017. https://research.google.com/audioset/

[9] EchoGuide - James Dyson Award Project (2026). https://www.jamesdysonaward.org/en-US/2026/project/echoguide/

[10] NEXUS - James Dyson Award Project (2026). https://www.jamesdysonaward.org/en-US/2026/project/nexus/